

MAWLANA BHASHANI SCIENCE AND TECHNOLOGY UNIVERSITY

SANTOSH, TANGAIL-1902



DEPARTMENT OF INFORMATION AND COMMUNICATION TECHNOLOGY

Lab Report

Lab Report No : 09

Lab Report on : Introduction to AJAX and Dynamic Department Data Retrieval from JSON
Using Faculty Selection Without Page Reload

Course Title : Web Application Development Lab

Course Code : ICT-3204

Submitted By	Submitted To
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Introduction:

AJAX (Asynchronous JavaScript and XML) is a web development technique that allows web pages to update specific parts of their content asynchronously without reloading the entire page. This improves user experience by making applications faster and more interactive. In modern web applications, AJAX is commonly used with JSON (JavaScript Object Notation) to exchange data between the client and the server efficiently.

In this experiment, AJAX is used to dynamically retrieve department information from a JSON file based on a selected faculty. When a user selects a faculty from a dropdown menu, the corresponding department list is displayed instantly without refreshing the page.

Objective:

The objectives of this lab are:

- To understand the basic concept of AJAX.
- To learn how to use XMLHttpRequest for asynchronous data retrieval.
- To parse and process JSON data using JavaScript.
- To dynamically update HTML content based on user input without page reload.

Tools and Technologies Used:

- HTML5
- JavaScript
- AJAX (XMLHttpRequest)
- JSON
- Web Browser (Chrome/Firefox)

Methodology:

1. A dropdown menu is created using HTML to select different faculties.
2. When a faculty is selected, the loadDept() JavaScript function is triggered using the onchange event.
3. An AJAX request is sent to fetch data from the info.json file.
4. The JSON response is parsed and matched with the selected faculty.
5. The corresponding department names are dynamically displayed inside the webpage without reloading.

Code Description:

- **HTML File:** Contains the dropdown list and a placeholder <div> to display department names.

```
index.html U X  info.json U  script.js U
AJAX Dropdown Example > index.html > html > body
1  <!DOCTYPE html>
2  <html lang="en">
3  <head>
4      <meta charset="UTF-8">
5      <title>AJAX Concept</title>
6      <script src="script.js"></script>
7  </head>
8  <body>
9
10 <h2>Select Department</h2>
11
12 <select onchange="loadDept(this.value)">
13     <option value="">-- Select Faculty --</option>
14     <option value="Engineering Faculty">Engineering Faculty</option>
15     <option value="LifeScience Faculty">LifeScience Faculty</option>
16     <option value="Science Faculty">Science Faculty</option>
17     <option value="Arts Faculty">Arts Faculty</option>
18 </select>
19
20 <div id="output">Department Name:</div>
21
22 </body>
23 </html>
```

- **JSON File:** Stores faculty names and their corresponding department lists in a structured format.

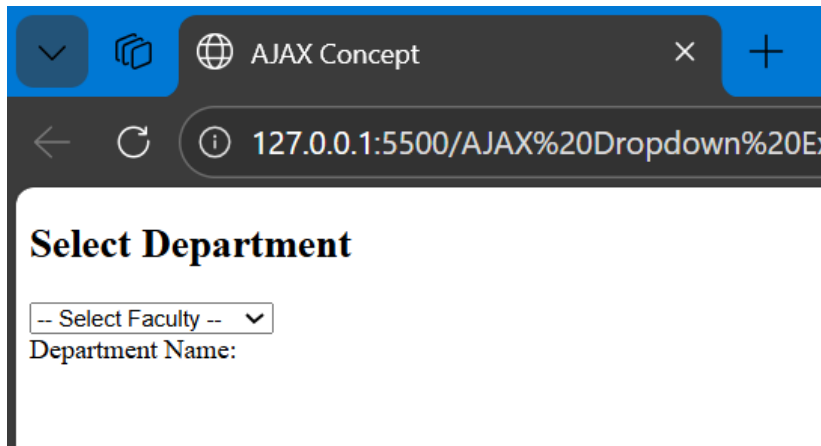
```
index.html U  info.json U X  script.js U
AJAX Dropdown Example > info.json > ...
1  [
2      { "faculty": "Engineering Faculty", "dept": ["ICT", "CSE", "ME", "TE"] },
3      { "faculty": "LifeScience Faculty", "dept": ["CPS", "ESRM", "SWE", "EEE"] }, "ESRM": Unknown wor
4      { "faculty": "Science Faculty", "dept": ["Math", "Physics", "Chemistry", "Statistics"] },
5      { "faculty": "Arts Faculty", "dept": ["Accounting", "Management", "English", "FTNS"] } "FTNS": U
6  ]
```

- **JavaScript File:** Uses XMLHttpRequest to fetch JSON data asynchronously and updates the DOM dynamically.

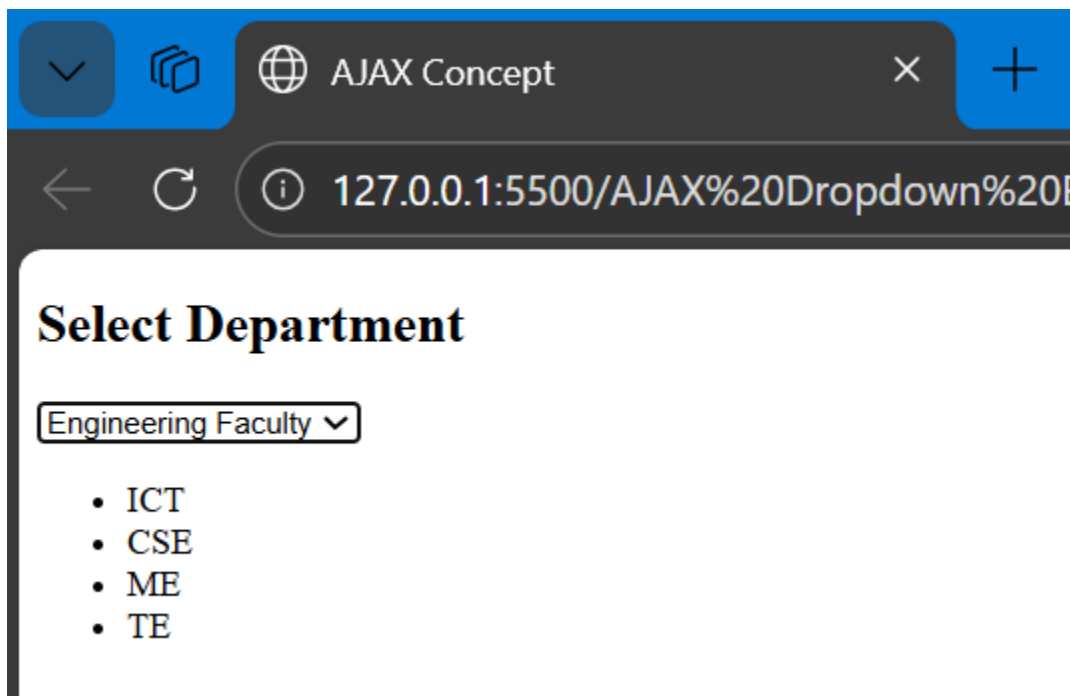
```
index.html U  info.json U  script.js U X
AJAX Dropdown Example > script.js > loadDept > onreadystatechange
1  function loadDept(faculty) {
2
3      if (faculty === "") {
4          document.getElementById("output").innerHTML = "";
5          return;
6      }
7
8      var xhr = new XMLHttpRequest();
9      xhr.open("GET", "info.json", true);
10
11     xhr.onreadystatechange = function () {
12         if (xhr.readyState === 4 && xhr.status === 200) {
13
14             var data = JSON.parse(xhr.responseText);
15             var html = "<ul>";
16
17             for (var i = 0; i < data.length; i++) {
18                 if (data[i].faculty === faculty) {
19                     for (var j = 0; j < data[i].dept.length; j++) {
20                         html += "<li>" + data[i].dept[j] + "</li>";
21                     }
22                 }
23             }
24
25             html += "</ul>";
26             document.getElementById("output").innerHTML = html;
27         }
28     };
29
30     xhr.send();
31 }
```

Output:

- Initially, the page displays a dropdown menu with faculty options.



- When a faculty (e.g., Engineering Faculty) is selected, the related departments (ICT, CSE, ME, TE) are displayed as a list.



- The page content updates instantly without any page reload.

Conclusion:

This lab successfully demonstrates the use of AJAX to retrieve and display data dynamically from a JSON file. It shows how asynchronous communication enhances web application performance and user experience. By implementing faculty-based department retrieval without page reload, the experiment highlights the practical use of AJAX and JSON in modern web development.